the other hand, the cost of wasting time may be paramount. A great tit when feeding nestlings needs to catch an average of one prey per thirty seconds. Every second of daylight is precious. Even the comparatively short time wasted in a hawk/hawk fight should perhaps be regarded as more serious than the risk of injury to such a bird. Unfortunately, we know too little at present to assign realistic numbers to the costs and benefits of various outcomes in nature.* We must be careful not to draw conclusions that result simply from our own arbitrary choice of numbers. The general conclusions which are important are that ESSs will tend to evolve, that an ESS is not the same as the optimum that could be achieved by a group conspiracy, and that common sense can be misleading.

Another kind of war game that Maynard Smith has considered is the 'war of attrition'. This can be thought of as arising in a species that never engages in dangerous combat, perhaps a well-armoured species in which injury is very unlikely. All disputes in this species are settled by conventional posturing. A contest always ends in one rival or the other backing down. To win, all you have to do is stand your ground and glare at the opponent until he finally turns tail. Obviously no animal can afford to spend infinite time threatening; there are important things to be done elsewhere. The resource he is competing for may be valuable, but it is not infinitely valuable. It is only worth so much time and, as at an auction sale, each individual is prepared to spend only so much on it. Time is the currency of this two-bidder auction.

Suppose all such individuals worked out in advance exactly how much time they thought a particular kind of resource, say a female, was worth. A mutant individual who was prepared to go on just a little bit longer would always win. So the strategy of maintaining a fixed bidding limit is unstable. Even if the value of the resource can be very finely estimated, and all individuals bid exactly the right value, the strategy is unstable. Any two individuals bidding according to this maximum strategy would give up at exactly the same instant, and neither would get the resource! It would then pay an individual to give up right at the start rather than waste any time in contests at all. The important difference between the war of attrition and a real auction sale is, after all, that in the war of attrition *both* contestants pay the price but only one of them gets the goods. In a population of maximum bidders, therefore, a strategy of giving up at the beginning would be successful and would spread